

FARIBAULT, MN, USA

WMO: 726563

Lat: 44.333N

Lon: 93.317W

Elev: 323

StdP: 97.50

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.3	(c) -22.5	(d) -29.0	(e) 0.3	(f) -24.2	(g) -27.1	(h) 0.3	(i) -22.3	(j) 11.1	(k) -9.0	(l) 10.1	(m) -9.3	(n) 2.2	(o) 290	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.3	23.4	30.9	22.7	28.8	21.6	25.4	30.2	24.2	28.5	23.1	27.2	4.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.4	28.5	22.6	18.1	27.1	22.1	17.4	26.4	79.7	30.0	74.5	28.7	70.2	27.4	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.2	(b) 8.8	(c) 8.0	(e) -29.0	(f) 34.9	(g) 4.5	(h) 1.4	(i) -32.3	(j) 36.0	(k) -34.9	(l) 36.8	(m) -37.5	(n) 37.6	(o) -40.7	(p) 38.6		
(a) 10.2	(b) 8.8	(c) 8.0	(e) -29.1	(f) 27.3	(g) 4.3	(h) 1.3	(i) -32.1	(j) 28.3	(k) -34.6	(l) 29.0	(m) -37.0	(n) 29.8	(o) -40.1	(p) 30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-9.5	-7.2	0.3	8.2	14.7	20.2	22.3	20.7	16.8	9.4	1.5	-6.2
	DBStd	12.39	7.35	7.07	6.94	5.54	4.92	3.65	3.07	3.04	4.73	5.31	5.99	6.66
	HDD10.0	2351	604	482	309	97	13	0	0	0	4	77	262	502
	HDD18.3	4249	863	715	559	307	134	22	4	12	84	282	506	760
	CDD10.0	1501	0	0	8	43	159	306	381	332	209	57	6	0
	CDD18.3	356	0	0	1	2	22	78	127	85	38	4	0	0
	CDH23.3	3260	0	0	2	33	256	702	1194	684	348	41	1	0
	CDH26.7	982	0	0	0	9	85	219	397	169	95	7	0	0
Wind	WSAvg	3.4	3.6	3.7	3.9	4.2	3.6	3.1	2.6	2.4	3.0	3.7	3.8	3.5
Precipitation	PrecAvg	857	25	21	51	72	103	128	111	124	82	63	44	29
	PrecMax	1037	69	49	96	142	205	230	224	262	255	147	129	94
	PrecMin	582	6	2	8	16	35	57	35	37	19	13	3	6
	PrecStd	118	16	13	24	33	47	46	50	50	53	38	33	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	10.2	22.0	27.5	32.6	33.0	34.0	32.4	32.2	27.7	20.1	12.2
		MCWB	4.0	7.4	14.3	16.7	19.9	23.3	25.6	24.7	22.4	17.5	13.1	9.9
	2%	DB	3.7	7.1	17.1	22.8	28.8	31.3	32.4	30.1	29.0	23.8	16.2	7.3
		MCWB	1.7	4.8	12.4	13.9	18.1	21.7	24.7	23.6	21.3	15.9	11.1	5.2
	5%	DB	2.2	4.0	13.0	20.2	26.3	29.0	30.9	28.0	27.4	21.1	12.7	4.0
		MCWB	0.9	2.1	9.5	12.2	16.8	20.8	23.8	21.9	20.0	14.8	9.0	2.2
	10%	DB	0.2	2.4	10.0	17.5	22.9	27.4	28.8	27.3	25.1	17.8	9.9	2.4
		MCWB	-0.5	1.1	7.2	10.9	15.1	19.7	22.5	21.4	18.7	13.0	6.7	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	7.7	15.8	18.1	22.2	25.6	27.6	26.2	24.6	20.2	14.9	9.7
		MCDB	6.5	9.6	19.5	24.1	29.0	31.8	32.3	30.7	29.4	24.8	18.8	11.3
	2%	WB	2.1	4.8	13.1	15.4	20.0	23.9	25.8	24.7	23.0	17.5	11.9	5.1
		MCDB	3.5	6.8	15.6	21.3	25.9	28.6	30.5	28.6	27.5	20.9	15.0	7.3
	5%	WB	0.8	2.6	10.2	13.5	18.5	22.4	24.6	23.5	21.4	15.6	9.4	2.8
		MCDB	2.0	4.1	13.3	18.5	23.7	27.3	29.2	27.1	25.4	19.6	12.7	4.1
	10%	WB	-0.5	1.1	7.2	11.8	17.0	21.1	23.5	22.4	19.9	13.7	7.2	1.2
		MCDB	0.2	2.4	9.8	16.4	21.5	25.6	27.6	26.0	23.8	17.7	9.4	2.4
Mean Daily Temperature Range		MDBR	9.2	9.4	9.7	11.1	11.4	10.9	11.1	10.9	12.0	10.8	8.9	8.1
	5% DB	MCDBR	10.0	9.6	13.1	15.8	15.5	13.3	12.5	11.9	13.8	14.9	13.3	9.9
		MCWBR	8.4	7.9	8.9	8.6	7.3	6.0	6.4	6.5	7.0	8.2	8.8	8.1
	5% WB	MCDBR	9.3	9.0	12.4	13.5	12.4	11.2	11.1	10.5	11.6	12.1	12.4	9.1
		MCWBR	8.1	7.5	8.9	8.5	6.7	6.5	6.5	6.6	6.9	7.8	8.7	7.6
Clear-Sky Solar Irradiance	taub	0.269	0.288	0.317	0.371	0.398	0.405	0.406	0.403	0.374	0.323	0.292	0.268	
	taud	2.426	2.354	2.401	2.303	2.284	2.305	2.308	2.315	2.365	2.501	2.502	2.445	
	Ebn at Noon	867	905	922	891	873	865	859	850	847	857	833	831	
	Edh at Noon	72	93	102	123	129	127	126	120	106	81	67	64	
All-Sky Solar Radiation	RadAvg	1.75	2.83	3.71	4.57	5.35	5.98	6.34	5.31	4.18	2.69	1.74	1.26	
	RadStd	0.21	0.30	0.37	0.49	0.43	0.49	0.26	0.37	0.31	0.36	0.17	0.12	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page